

RC Car Level 3 Validation

Advanced Autonomy & Robustness Testing

1 Basic Stability & Reliability

Goal: Ensure the core system is rock-solid before complex testing.

- ☐ **Endurance:** Runs for 5+ minutes continuously without freezes or manual resets.
- ☐ **Control Quality:** Steering and throttle transitions are fluid (no jitter).
- ☐ **Latency:** Zero significant lag spikes under high processing load.

Metric	Pass	Partial	Fail
Current Status	☐	☐	☐

Notes: _____

2 Environmental Robustness

Goal: High performance across varying real-world conditions.

- ☐ **Lighting:** Reliable performance in bright sun, harsh shadows, and low-light dusk.
- ☐ **Traction:** Consistent handling on asphalt, concrete, and light gravel.
- ☐ **Vibration:** Camera/LiDAR mounts remain stable; code handles minor angle shifts.

Metric	Pass	Partial	Fail
Current Status	☐	☐	☐

Notes: _____

3 Static Obstacle Handling

Goal: Spatial awareness and precision avoidance.

- ☐ **Thin Objects:** Detects and avoids poles, table legs, or thin sticks.
- ☐ **Low Profile:** Correctly identifies curbs, rocks, or steps below the main sensor line.
- ☐ **Proximity:** Vehicle executes a controlled emergency stop if path is fully blocked.

Metric	Pass	Partial	Fail
Current Status	☐	☐	☐

4 Dynamic Obstacles & Safety

Goal: Interaction with a changing world.

- ☐ **Moving Targets:** Predicts and avoids moving people, pets, or other RC cars.
- ☐ **Reaction Speed:** Braking or swerving happens smoothly without losing traction.
- ☐ **Recovery:** Resumes mission path immediately after the obstacle clears.

Metric	Pass	Partial	Fail
Current Status	☐	☐	☐

5 Sensor Fusion & Logic Conflict

Goal: Smart decision making when data disagrees.

- ☐ **Priority Logic:** LiDAR overrides camera when distance data detects a collision.
- ☐ **Confidence Thresholds:** System slows down if camera vision confidence drops below 70%.

Metric	Pass	Partial	Fail
Current Status	☐	☐	☐

Final Readiness Score

Total Sections Passed	/ 5	
Level 3 Deployment Ready?	☐ YES	☐ NO